Substructures identification and mass estimation

For the subhalo statistics study using mass maps derived from lensing and U-Net models for the CLASH clusters, we developed a subhalo identification and mass estimation algorithm. The aim is to identify subhalos directly from the mass maps and to define a criterion for assigning an adaptive aperture for their two-dimensional mass estimation.

We begin by first addressing the problem of aperture estimation for subhalos when their locations are known *apriori*. For this, we used the available AHF halo catalog from the The300 simulation. We considered the training-set mass maps constructed with the line of sight along the Z-axis, and applied the same selection to the AHF catalog used in generating the projected mass maps along this direction, in order to identify the corresponding subhalos in the maps. We selected all subhalos gravitationally bound to the main host halo within the field of view, along with any additional objects falling within the same projected field of view and line-of-sight depth, in order to account for projected structures along the line of sight.

C.1. Adaptive Aperture algorithm for Substructures

For an identified subhalo with peak position $p_i = (x_i, y_i)$ in the two-dimensional mass map, we define an adaptive aperture radius $R_{i,\mathrm{ap}}$ centered on the peak. The aperture is calculated to scale with the relative prominence of the peak while also account for both local background variations and spatial distribution of neighboring subhalos.

We define a reference surface density at each peak as a weighted combination of a local background estimate and a global mean over all peaks,

$$M_{i,\mathrm{ref}} = w\, M_{i,\mathrm{local}} + (1 - w)\, \bar{M}_{\mathrm{sub}}, \tag{C.4}$$

where $M_{i,\mathrm{local}}$ is the local background surface density around the peak, $\bar{M}_{\mathrm{sub}}$ is the mean surface density evaluated at all peak locations, and we adopt a fixed weight $w = 0.5$. The aperture scale is defined as

$$R_{i,\mathrm{phys}} = \alpha \left(\frac{M_{i,\mathrm{peak}}}{M_{i,\mathrm{ref}}} \right)^{\gamma} \frac{d_{\mathrm{nn}}}{2}, \tag{C.5}$$

where $M_{i,\mathrm{peak}}$ is the projected surface density at the peak and d_{nn} is the distance to the nearest neighboring peak. To avoid nonphysical small apertures, we introduce a smooth minimum scale R_β, such that the final aperture radius is given by

$$R_{i,\mathrm{ap}} = \begin{cases} \sqrt{R_{i,\mathrm{phys}}^2 + R_\beta^2}, & \text{if } R_{i,\mathrm{phys}} < R_\beta, \\ R_{i,\mathrm{phys}}, & \text{otherwise.} \end{cases} \tag{C.6}$$

The above procedure that subhalos in dense environments are assigned apertures relative to their local background, while also controlling excessive overlap through the dependence on the nearest-neighbor distance. The parameter γ controls the sensitivity of the aperture size to the peak contrast, and R_β acts as a regularizer. The enclosed mass is computed by integrating the surface density within a circular aperture of radius $R_{i,\mathrm{ap}}$. To achieve subpixel accuracy, the mass map is interpolated onto a high-resolution grid within the aperture using interpolation, and the integral is evaluated numerically over the selected pixels. The free parameters $\{\alpha, \gamma, R_\beta\}$ are calibrated on a training set of 500 simulated cluster mass maps drawn from the full sample of 3273 clusters. We employ a Bayesian optimization scheme to determine the parameter combination that best reproduces the known three-dimensional subhalo masses M_{500c} from the AHF catalog, by jointly matching the differential and cumulative subhalo mass functions. This procedure yields the parameters values $\alpha = 0.21, \gamma = 1.2,$ and $R_\beta = 0.95$.

The performance of the adaptive aperture algorithm is illustrated in Fig. C.2, where we validate the method using a subset of 1200 cluster mass maps drawn from the The300 simulation, under the assumption that the true subhalo positions are known *apriori* from the AHF catalog. In the top panels, we present a visual representation of the method, showing two representative projected mass maps with the corresponding adaptive apertures overlaid. In the lower panels, we compare the subhalo mass functions derived from the two-dimensional aperture masses with those obtained from the three-dimensional AHF catalog. The cumulative and differential subhalo mass functions are shown, with shaded regions indicating the 16th–84th percentile range across the full sample, and solid lines representing the median relations. We find good agreement between the two mass functions, particularly at the massive end, in both shape and normalization. This demonstrates that the calibrated aperture prescription successfully recovers the underlying subhalo mass function, indicating that the algorithm is well tuned.

C.2. Substructures Identification from Mass Maps

To identify substructures directly from projected mass maps, we developed a multistage substructure detection algorithm designed to robustly extract density peaks across range of spatial scales. The algorithm mainly uses the following three steps for making detection:

1. Background estimation and standardization: We construct a radially varying background for each mass map in order to isolate local density fluctuations associated with potential substructures by removing the overall smooth cluster component. This is achieved using a combination of median filtering and Gaussian smoothing at different spatial scales. Specifically, we first apply two median filters: a fine-scale filter with a kernel size proportional to the base smoothing scale σ_0, and a coarse-scale filter with twice this kernel size. Each of these maps is then smoothed with Gaussian kernels of widths σ_0 and $1.5\,\sigma_0$, respectively. To account for radial variations in the cluster profile, the two background estimates are smoothly blended using a sigmoid

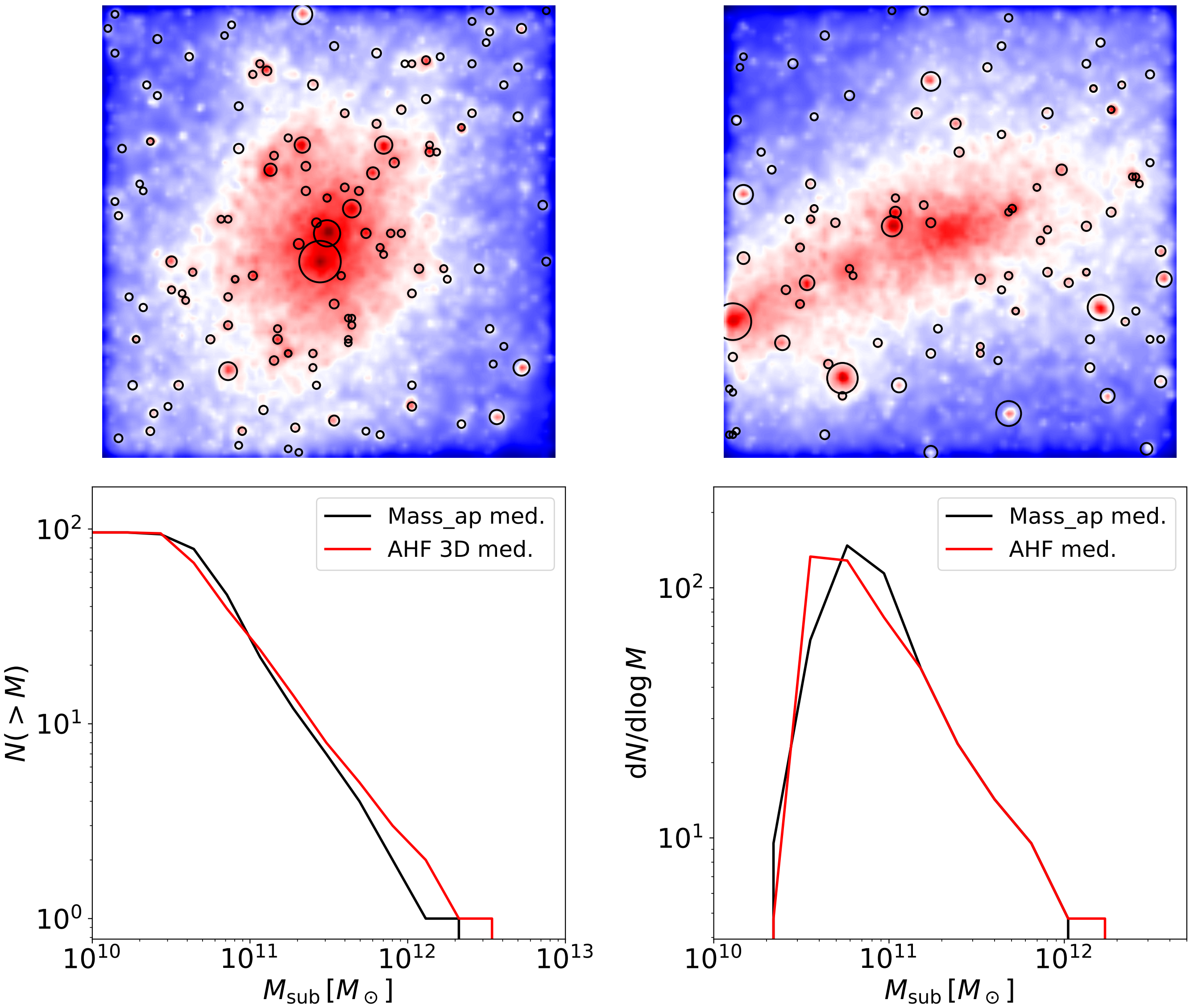

Fig. C.2: **Validation of the adaptive aperture algorithm**. Top panels: two representative projected mass maps with adaptive apertures overlaid as black circles when the subhalos are known a priory. Bottom-left: cumulative subhalo mass function $N(> M)$ computed from the 2D aperture masses (grey envelope and black median), compared to the 3D AHF catalog masses (red envelope and red median), both shown as the 16th–84th percentile range across all 3273 simulated clusters. Bottom-right: corresponding differential subhalo mass function.

weighting function, defined as $w_{\rm outer} = 1/\left(1 + \exp\left[-\frac{r - r_{\rm mid}}{r_{\rm width}}\right]\right)$. The spatial radial scale $r_{\rm mid}$ and $r_{\rm width}$ controls the transition from fine background near the center to a much coarser background at the outskirts for the mass map of the cluster. The final background is then given by $M_{\rm bkg} = (1 - w_{\rm outer})\, M_{\rm inner} + w_{\rm outer}\, M_{\rm outer}$. The resulting background map is subtracted from the original mass map to produce a residual map that highlights local over-densities. This residual map is subsequently standardized by subtracting its mean and dividing by its 1σ standard deviation, and is used as input for the peak detection stage.

2. Multi-scale peak detection : For the next step, we employ a multiscale detection scheme on the background-subtracted and standardized map. The standardized map is smoothed at different spatial scales using a sequence of logarithmically spaced Gaussian kernels, defined as $\sigma_s = \sigma_0 \{^s$, where $\{ = 1.25$ is the scale factor and s denotes the smoothing step, ranging from 0 to 5. This results in a total of six smoothing scales used for substructure detection. Peaks are initially detected at the finest smoothing scale, i.e. at σ_0, using a local maximum finder from `skimage`, with a minimum separation proportional to σ_0 and an absolute detection threshold $\alpha_{\rm det}$. The identified candidate peaks are then validated using the multiscale scheme. At each smoothing scale, the maximum value within a small neighborhood around each candidate peak is evaluated, allowing for slight positional drift. A peak is considered persistent at a given scale if its value remains above a fraction η of its base-scale amplitude. We assign a score of +1 whenever this condition is satisfied. A candidate is accepted as a genuine detection only if it persists in at least 50% of the smoothing scales. The motivation for this multi-scale scheme is to retain genuine detections while rejecting spurious

noise fluctuations, as true subhalo peaks persist across multiple smoothing scales, whereas noisy peaks are washed out. The free parameters smoothing scale σ_0, the detection threshold $\alpha_{\rm det}$, and the persistence threshold η, are constrained through calibration with the AHF catalog.

3. Signal-to-noise (SNR) filtering : To further suppress spurious detections, we compute a local SNR ratio for each candidate peak. The noise is estimated from an annular region centered on the peak, with inner and outer radii adapted to the distance to the nearest neighboring peak in order to avoid contamination from nearby structures. Specifically, the noise is estimated as $1.4826 \times$ the median absolute deviation within the annulus, and the SNR was computed using this noise, mass at the peak value and mass within the annular region. A potential substructure is retained if it satisfies the afore-mentioned persistence criterion and minimum SNR threshold of α_{snr}. The final output of the algorithm is the set of detected substructures position in the two-dimensional mass map.

The hyperparameters $\{\sigma_0, \alpha_{\rm det}, \eta, \alpha_{\rm snr}\}$ are optimized using simulated cluster mass maps, where the true substructures (halos) are taken from the AHF catalog. We use `Hyperopt` with Bayesian optimization (TPE algorithm) to explore the parameter space. For each set of parameters, the detection algorithm is run on the small sample of pre-selected mass maps, and the predicted subhalos are compared with the AHF halos. The performance is evaluated using a combined loss that includes agreement with the true mass function ($dN/d\log M$), along with precision and recall. The optimizer then tries new parameter values, focusing more on those that give lower loss. In this way, the algorithm is tuned to recover both the correct number of subhalos and their positions as given by AHF. The Bayesian optimization constrained these hyper-parameters values to $\sigma_0 = 0.75, \alpha_{\rm det} = 1.01, \eta = 0.32$, and $\alpha_{\rm snr} = 1.90$..These parameters balance completeness and precision in the detected subhalo population.

When this hyper-parameter peak detection algorithm is evaluated on 2000 randomly selected maps from the The300 sample, it results in a median recall of 0.750 ± 0.149 and a median precision of 0.881 ± 0.102. The recall measures the fraction of true AHF subhalo peaks that are successfully recovered, while precision quantifies the fraction of detected peaks that correspond to genuine subhalos. This implies that approximately 25 per cent of the true subhalo peaks are missed by the algorithm, while about 12 per cent of the detections are false positives. Restricting to subhalos above the resolution limit of $10^{11}, M_\odot$, the algorithm achieves an average completeness of 0.952 ± 0.060, requiring detections to lie within 2 pixels ($\approx$ 8 kpc) of the true AHF positions. Overall, this demonstrates that the calibrated peak detection algorithm reliably recovers the underlying subhalo population, particularly at the high-mass end, and supports its application to both simulated data and observed CLASH cluster mass maps

Fig. C.3 presents the performance of the calibrated peak detection algorithm evaluated on the full The300 simulation sample. The top panels show two representative projected mass maps, with the true AHF subhalo positions and the algorithm-detected peaks overlaid. The algorithm successfully recovers the majority of true subhalo locations, with only a small number of missed detections and false positives, mainly in the dense central regions. The bottom panels show the cumulative and differential subhalo mass functions derived from the aperture masses of the detected peaks (Algo-2D) and the true AHF peaks (AHF-2D), shown as the median and 16th–84th percentile range across the full sample. The two mass functions are in good agreement in both shape and normalization, with the median line lying within scatter over the full subhalos mass range. In Fig. C.4, we show the identified substructures and their apertures in the lensing maps (top panel) and the multi-encoder U-Net predicted maps (bottom panel) for the five main CLASH clusters discussed in Section 6 of the main manuscript.

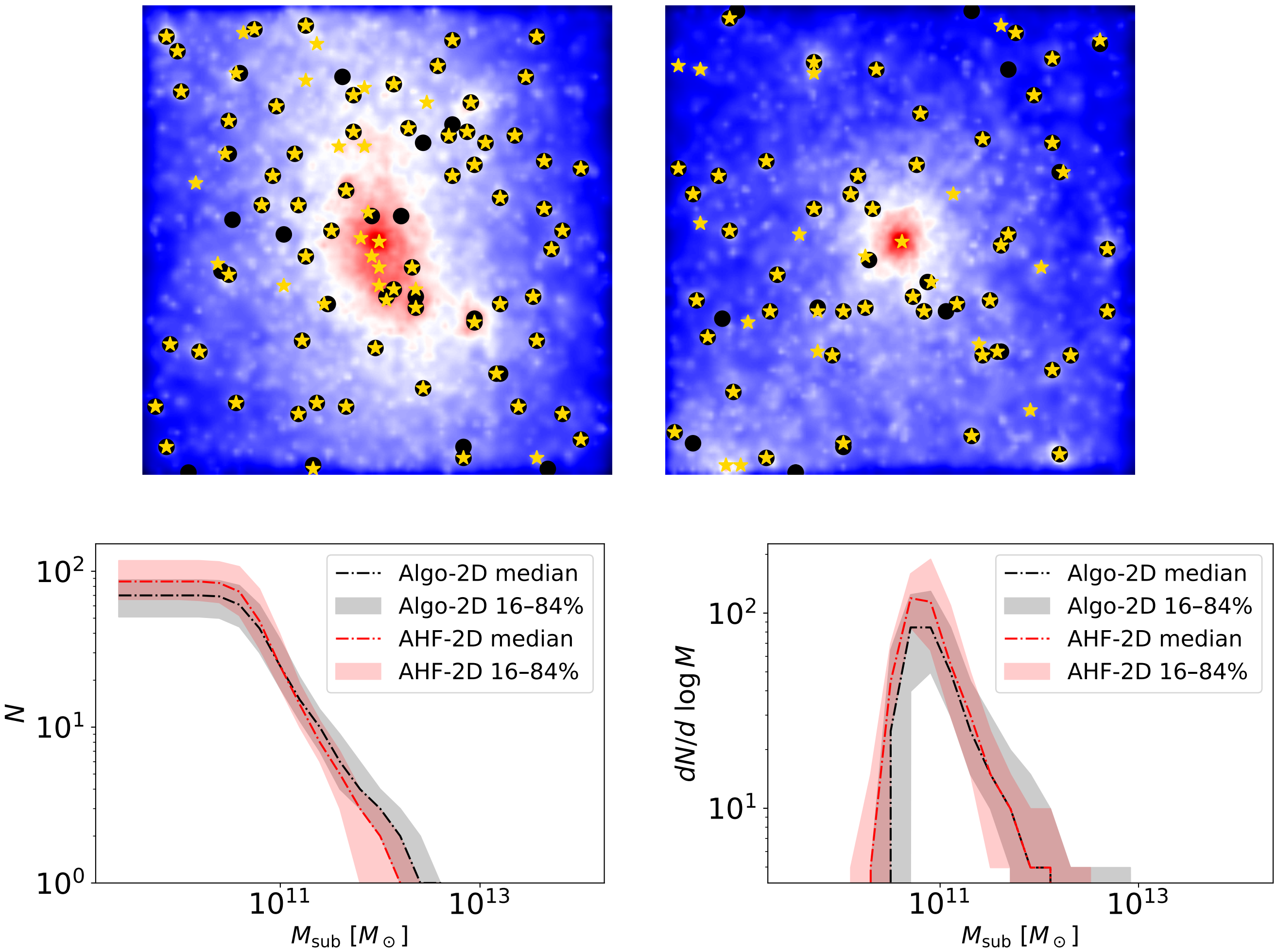

Fig. C.3: **Subhalo peak detection algorithm performance** Top panels: two representative projected mass maps from the test set, with true AHF subhalo positions marked as black circles and algorithm-detected peaks as gold stars. The algorithm recovers the majority of true subhalo positions, including substructures in the outskirts of the cluster, with only a small number of missed detections and false positives. Bottom-left: cumulative subhalo mass function computed from the aperture masses of the algorithm-detected peaks (black, Algo-2D) and the true AHF peaks (red, AHF-2D), shown as the median and 16th–84th percentile range across the full simulation sample. Bottom-right: corresponding differential subhalo mass function.

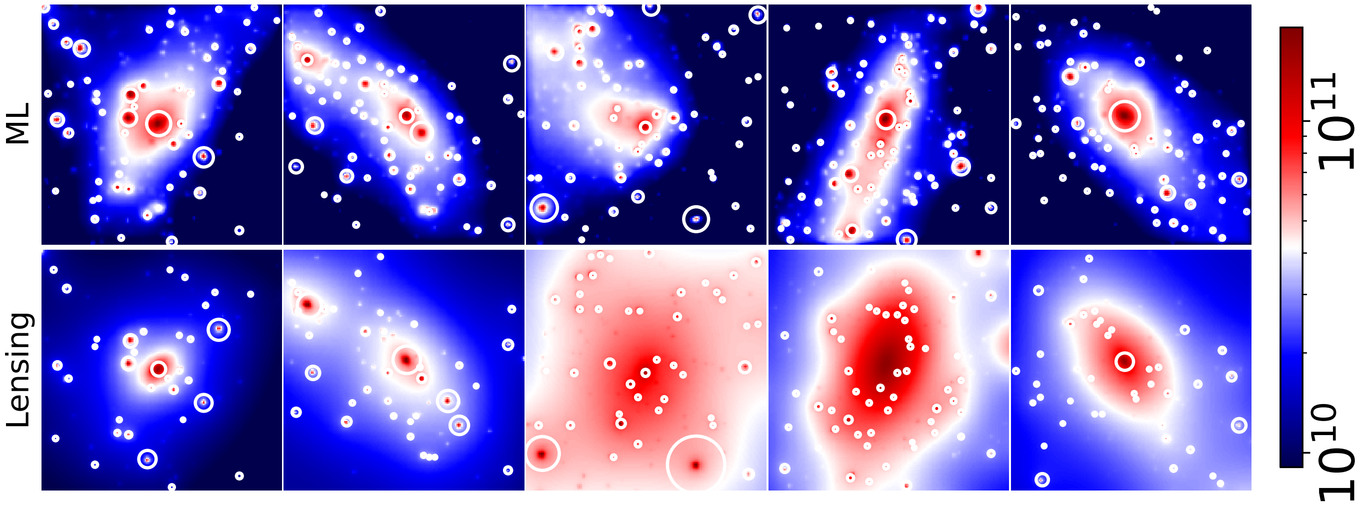

Fig. C.4: **Substructures in CLASH sample.** Projected mass maps of the five CLASH clusters with spectroscopic redshift catalogs, with identified subhalo peaks and their adaptive apertures overlaid as white circles. The upper panels show the lensing-derived mass maps and the lower panels show the corresponding U-Net-predicted mass maps. The radius of each identified substructure is given by (Eq. C.6).